Detecting Political Affiliation in French Electoral Manifestos (1973–1993)

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Abstract

This work examines to what extent the rhetorical style of electoral manifestos can predict a candidate's political affiliation in French legislative elections of the Fifth Republic (1973–1993), independently of any explicit mention of their party. Drawing on the Archelec corpus (Sciences Po), we compare three classification approaches of increasing complexity: TF-IDF with Naive Bayes, TF-IDF with linear SVM, and CamemBERT in zero-shot mode. To isolate pure rhetorical fingerprints, we conduct two parallel experiments: one on raw text and one on text from which party names have been removed. The SVM achieves 97% accuracy in both experiments, confirming that electoral manifestos encode a rhetorical signature specific to each political family.

1 Introduction

Electoral manifestos constitute privileged material for the analysis of political discourse. Unlike interviews or public speeches, they are written freely by each candidate and directly reflect their rhetorical and thematic choices. The Archelec corpus, compiled and digitized by Sciences Po, gathers 21,171 electoral manifestos from candidates in French legislative elections of the Fifth Republic between 1973 and 1993.

This work addresses the following question: can we predict a candidate's political affiliation from the text of their electoral manifesto alone? This question raises a central methodological issue: 79% of manifestos explicitly mention the name of the candidate's party. We thus ask to what extent a model learns rhetorical style rather than simply reading the party name. To answer this question, we conduct two parallel experiments. Experiment 1 uses raw text; Experiment 2 uses text from which party names have been removed. The difference in performance between the two measures to what extent political affiliation is explicitly declared rather than implicitly encoded in the discourse.

2 Data and Preprocessing

The Archelec corpus covers five legislative elections: 1973, 1978, 1981, 1988 and 1993. The documents were digitized by OCR from paper documents, which introduces typical artifacts: misrecognized characters, repeated watermarks, multiple spaces. The metadata associated with each manifesto — political party, department, profession and sex of the candidate — come from the Sciences Po Archelec portal and were merged with the text files to constitute our working corpus.

To build an exploitable corpus, we retain only parties with at least 50 candidates, guaranteeing a minimal volume of training examples. The 37 parties thus retained are grouped into 6 political families according to the left-right positioning commonly accepted for the Fifth Republic. After merging with the text files, the working corpus comprises 12,613 documents distributed as follows:

Table 1: Distribution of the corpus by political family.

Political family	Documents	Proportion
Left	5,450	43.2%
Right	2,495	19.8%
Far right	1,537	12.2%
Far left	1,411	11.2%
Ecologists	1,031	8.2%
Centre	689	5.5%
Total	**12,613**	**100%**

Figure 1 illustrates this distribution and its evolution by electoral year. A peak is observed in 1993, particularly pronounced on the right, resulting from a wave of rejection following Mitterrand's second term begun in 1988.

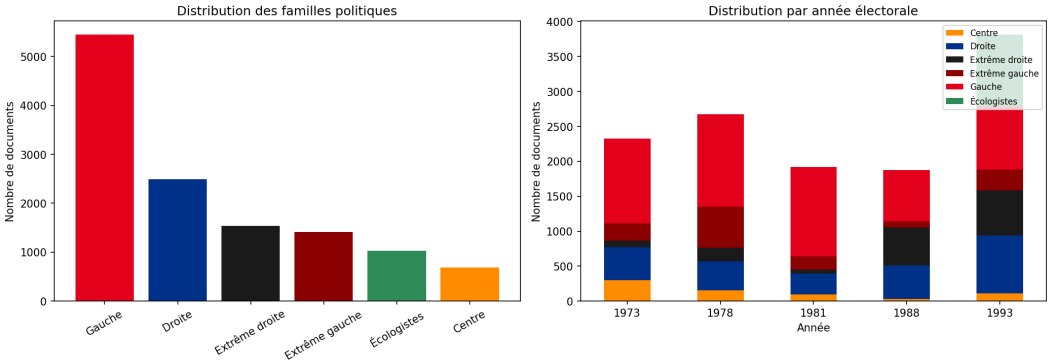

Figure 1: Distribution of political families and evolution by electoral year.

The cleaning pipeline applied to the texts comprises the removal of OCR artifacts, lowercasing, removal of punctuation and digits, followed by lemmatization with spaCy (`fr_core_news_sm`) and stopword removal. This pipeline is applied to both versions of the corpus. Since 79% of manifestos explicitly mention the party name, party names are removed before lemmatization for Experiment 2 using regular expressions. A post-removal verification confirms that the mention rate drops from 79% to 0.2%.

The analysis of manifesto length (Figure 2) reveals distinct patterns across families. The Far left presents the longest texts (mean 1,220 words) with very low variance — the box is both high and narrow — which contrasts sharply with the other families. Ecologists produce the shortest texts (468 words) with high dispersion. This heterogeneity between families, and notably the remarkable homogeneity of the Far left, is certainly the reflection of differentiated internal organization: some parties centrally produce a standardized manifesto template, while others leave more freedom to their local candidates.

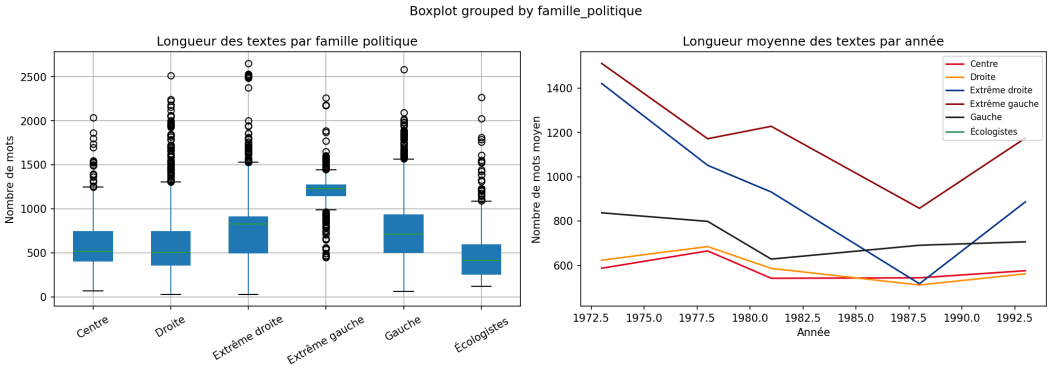

Figure 2: Length of electoral manifestos by political family and temporal evolution.

3 Methods

The corpus is split into a training set (90%) and a test set (10%) with stratification on political families. Each model is trained twice: on `text_processed` (Experiment 1, with party names) and on `text_no_party_processed` (Experiment 2, without party names).

3.1 TF-IDF + Naive Bayes

Each manifesto is first transformed into a numerical vector by the TF-IDF representation. For each term t of the vocabulary, the TF-IDF score combines its frequency in the document with an IDF factor that penalizes it if it is heavily present across the corpus — words that are frequent everywhere carry little discriminative information. The score is defined by:

$$\text{TF-IDF}(t, d, D) = \frac{f_{t,d}}{\sum_{t'} f_{t',d}} \times \log \frac{|D|}{|\{d \in D : t \in d\}|} \tag{1}$$

We use 5,000 features with unigrams and bigrams (`sublinear_tf=True`). The multinomial Naive Bayes classifier then exploits these vectors by assuming the conditional independence of words given the class. It predicts the political family with maximum likelihood:

$$\hat{c} = \arg\max_{c \in \mathcal{C}} \left[\log P(c) + \sum_{i=1}^{n} \log P(w_i|c) \right] \tag{2}$$

Table 2: Naive Bayes results on the test set.

Experiment	Accuracy	F1-macro
Exp1 (with party names)	0.902	0.843
Exp2 (without party names)	0.893	0.829

3.2 TF-IDF + Linear SVM

We build on the same TF-IDF vectorization. Unlike Naive Bayes, which models the distribution of words per class, the SVM directly seeks a separating hyperplane in this high-dimensional space. It maximizes the margin between classes, making it particularly robust when the number of features exceeds the number of examples — a frequent situation in text classification. The objective function is:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \max(0, 1 - y_i(w \cdot x_i + b)) \tag{3}$$

The parameter C controls the trade-off between the margin and classification errors. We use `class_weight=balanced` to compensate for class imbalance.

Table 3: Linear SVM results on the test set.

Experiment	Accuracy	F1-macro
Exp1 (with party names)	0.975	0.961
Exp2 (without party names)	0.968	0.949

3.3 CamemBERT Zero-Shot

We finally evaluate a zero-shot approach with the model `cmarkea/distilcamembert-base-nli`, a language model pre-trained on French. Unlike the two previous approaches, no supervised training is required. The model projects the text and the six political family labels ("left", "right", "far left", "far right", "ecologists", "centre") into a shared semantic space, then predicts the family whose label is semantically closest to the text. This approach is only evaluated on Experiment 2, to test whether the model recognizes political families from rhetorical style alone.

Table 4: CamemBERT Zero-Shot results on the test set (Exp2).

Model	Accuracy	F1-macro
CamemBERT Zero-Shot	0.208	0.216

4 Results and Discussion

The SVM largely outperforms Naive Bayes (97% vs. 90% accuracy), confirming the advantage of margin-based models in high-dimensional spaces. CamemBERT Zero-Shot (21%) falls well below the supervised approaches, indicating that detecting political affiliation in this historical corpus requires specific supervised learning. The model pre-trained on modern French struggles to capture the political vocabulary of the 1973–1993 period.

The most salient result concerns the comparison between the two experiments. The SVM loses only 0.7 accuracy points when moving from Experiment 1 to Experiment 2 (97.5% $\rightarrow$ 96.8%). This result demonstrates that electoral manifestos encode a rhetorical signature specific to each political family, independently of explicit party mentions. Candidates do not simply name their party; they adopt a vocabulary, themes and style that identify them.

The confusion matrices (Figures 3, 4 and 5) allow us to go beyond global metrics. For both Naive Bayes and SVM, the Centre is systematically confused with the Right. This confusion is not a failure of the model but a validation of its relevance: the UDF voted predominantly with the RPR between 1973 and 1993, and the two families effectively share a close rhetorical space. Similarly, the residual confusions between Far left and Left are politically coherent. For CamemBERT Zero-Shot, the model massively predicts the Centre class for the entire corpus, revealing a total inability to discriminate between political families in this historical context.

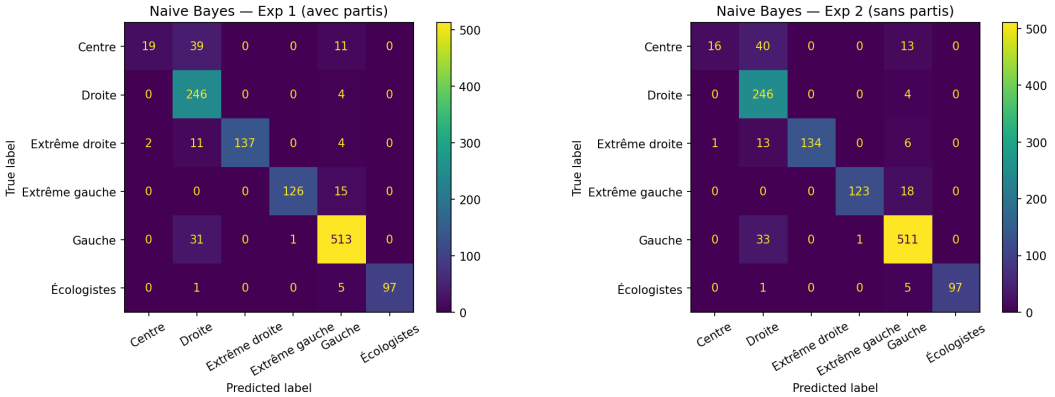

Figure 3: Confusion matrices of Naive Bayes, Experiment 1 (left) and Experiment 2 (right).

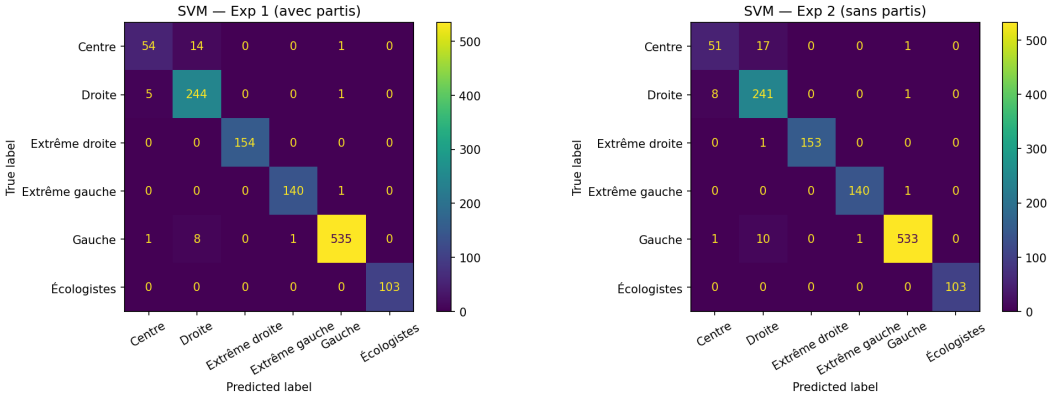

Figure 4: Confusion matrices of the SVM, Experiment 1 (left) and Experiment 2 (right).

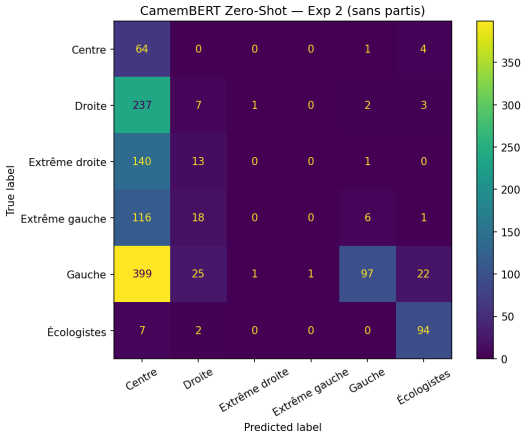

Figure 5: Confusion matrix of CamemBERT Zero-Shot (Experiment 2).

To deepen our understanding of the results, we analyze the coefficients of the SVM model trained on Experiment 2. These coefficients identify the words whose presence is most decisive for predicting each political family, independently of any party name. Figure 6 reveals clear lexical signatures: the

Far right is characterized by the terms “immigré” (immigrant), “immigration”, “patrie” (homeland) and “nation”; the Far left by “travailleur” (worker), “patron” (employer) and “mobilisation”; Ecologists by “nature”, “écologie” and “vert” (green). These results confirm that each political family develops a vocabulary consistent with its ideological positions.

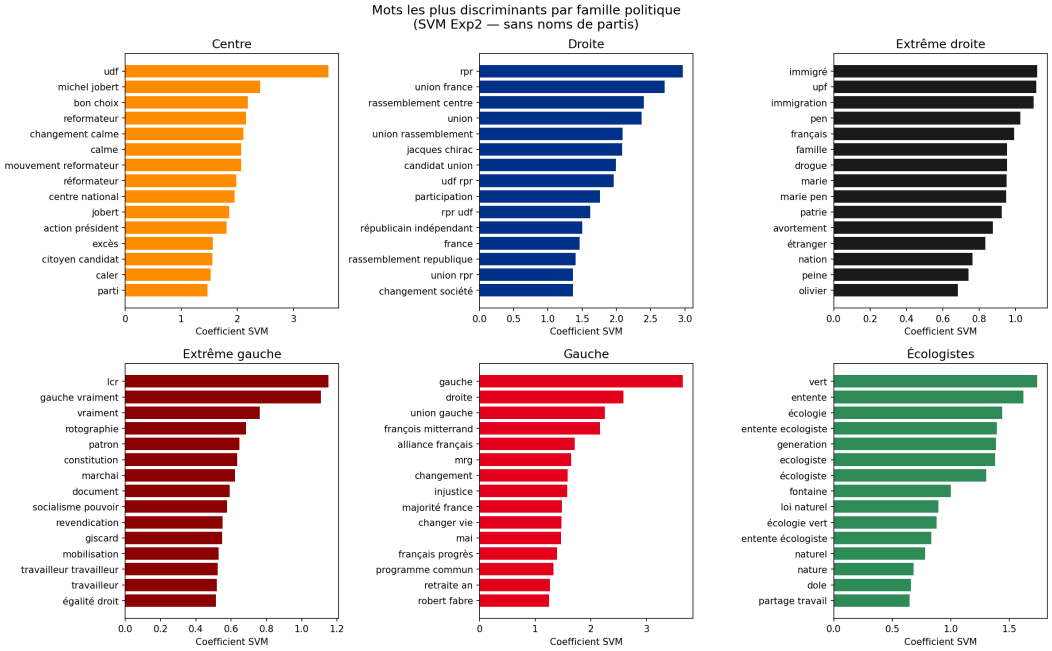

Figure 6: Most discriminant words by political family, based on SVM coefficients (Experiment 2).

The word clouds (Figure 7) and the tSNE projection (Figure 8) complement this analysis. The tSNE projection visually confirms the separability of political families: the Far right and Far left form compact clusters well isolated from the rest of the corpus, while the Centre merges geometrically with the Right, consistent with the confusions observed in the confusion matrices.

Figure 7: Word clouds by political family (without party names).

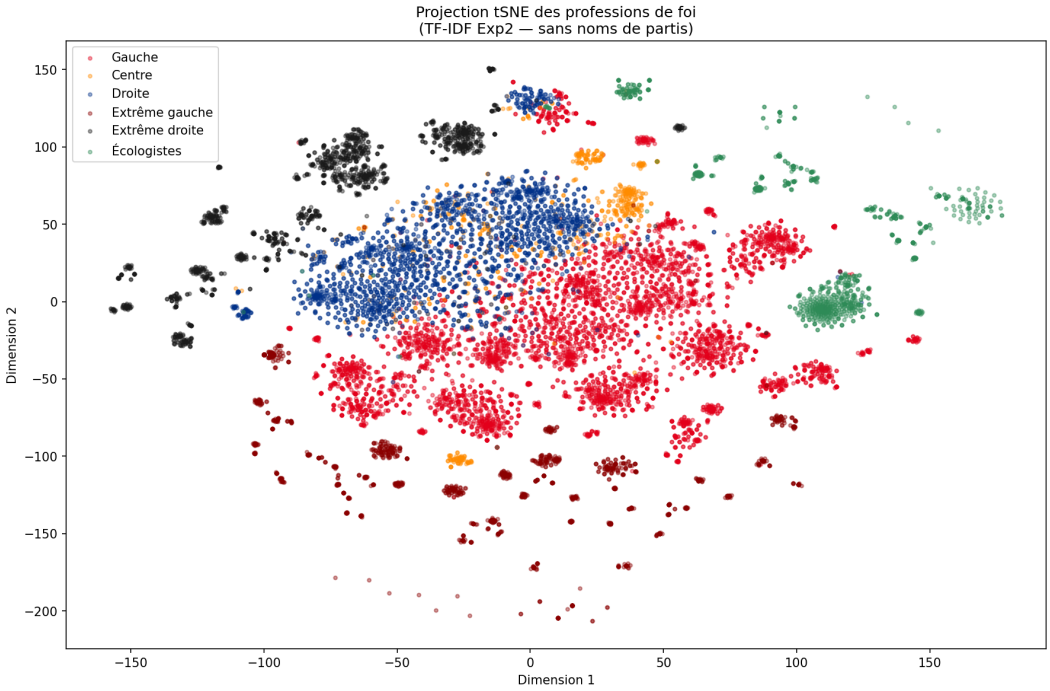

Figure 8: tSNE projection of TF-IDF representations (Experiment 2).

5 Conclusion

This work has demonstrated that it is possible to predict a candidate's political affiliation from the text of their electoral manifesto with 97% accuracy (SVM). More importantly, the removal of party names does not significantly degrade performance, confirming that each political family develops a distinctive and identifiable vocabulary and rhetorical style. The residual confusions are politically interpretable: the Centre and the Right share a close rhetorical space, consistent with their ideological proximity during the Fifth Republic. CamemBERT Zero-Shot fails to capture these historical nuances, underscoring the need for supervised learning for this type of corpus. Among the limitations of this work, we note that party abbreviations (RPR, UDF, LCR) were not removed in Experiment 2, and that class imbalance may favor predictions toward majority classes.